Quantitative Macro Theory **HW #5**
Spring 2019 **Due at 5pm on *Sunday* 3/3/2019**

Write your answer on latex and prepare a pdf file. The file should be clearly written and organized. You should think of that file as a short version of a paper. The appendix of that file should include your code. Email me that pdf file before the deadline.

Calibration:

- Individuals live between age $n = 25$ and $n = 80 = N$. The retire at $n = 65 = T$.
- $u(c) = \frac{c^{1-\sigma}}{(1-\sigma)}$, with $\sigma = 2$.
- The discount rate, $\beta = 0.92$.
- Interest rate, $r = 0.02$.
- Income is given by y, which has three components:

$$y = \exp(z + f(n) + \varepsilon)$$

- Use a three-state Markov chain Γ for the persistent component of income z, that we loosely identify with "blue-collar" ($z_1 = \log(15)$), "white-collar" ($z_2 = \log(80)$), and "rich" ($z_3 = \log(500)$).
- Transition matrix:

$$\begin{bmatrix} \Gamma_{1,1} = 0.60 & \Gamma_{1,2} = 0.35 & \Gamma_{1,3} = 0.05 \\ \Gamma_{2,1} = 0.10 & \Gamma_{2,2} = 0.80 & \Gamma_{2,3} = 0.10 \\ \Gamma_{3,1} = 0.05 & \Gamma_{3,2} = 0.35 & \Gamma_{3,3} = 0.60 \end{bmatrix}$$

- The deterministic component of earnings is estimated using the PSID data. It has a hump-shaped profile where earnings grow by approximately 60% during the first 25 years and then decrease approximately by 18% until the end of the working life.

$$f(n) = \log(-1.0135 + 0.1086 * n - 0.001122 * n^2).$$

- And ε takes two values, {log(0.8),log(1.2)}, each with probability 1/2.
- Finally, during retirement, which starts at n=66, their retirement income does not change and it is a function of the persistent component of income at age n=65:

$$R(z) = 0.7 \times \exp(z_{65} + f(65)).$$

- Set $\underline{a} = 0$.

In terms of the lifetime utility, we have that for age $n = 20$ to $n = 64$, the value is

$$V(a, z_i, \varepsilon, n) = \max_{c,a'} u(c) + \beta \sum_{j=1}^{3} \Gamma_{i,j} \left[0.5 * V(a', z_j, \varepsilon_L, n+1) + 0.5 * V(a', z_j, \varepsilon_H, n+1)\right]$$

subject to $c + a' = Ra + y(z_i, \varepsilon, n),\ a' \geq \underline{a}$.

Similarly, for age $n = 65$, we have

$$V(a, z_i, \varepsilon, 65) = \max_{c,a'} u(c) + \beta V(a', z_i, 66)$$

subject to $c + a' = Ra + y(z_i, \varepsilon, n),\ a' \geq \underline{a}$.

For age $n = 66$ to $n = 79$, consider the problem

$$V(a, z_i, n) = \max_{c,a'} u(c) + \beta V(a', z_i, n+1)$$

subject to $c + a' = Ra + R(z_i),\ a' \geq \underline{a}$.

Finally, for age $n = 80$, we have that $V(a, z_i, 80) = u(Ra + R(z_i))$.

1. (100 points) **Households life-cycle problem and stationary distribution**

   (a) (10 points) Solve the problem of an individual age 80. Then solve the problem of someone in retirement, backward from age 79 to 66. Using that value, solve the problem of an individual 65 years old. Now, we that value, solve the problem of an individual 64 years old and continue like that until age 20. Plot in the same plot $a'(a, z_1, \varepsilon_L, 25)$, $a'(a, z_1, \varepsilon_L, 40)$, and $a'(a, z_1, \varepsilon_L, 60)$. Use your preferred method.

   (b) (10 points) Simulate an individual starts with $a = 0$ and "blue-collar" worker income. Think that this individual is 20 years old. Simulate a path of income for this individual for 40 years, until she is 65 years old. Using that path, simulate the path of assets using $a'(a, y)$. Plot the simulated path of consumption, assets, and income in the same graph. In the x axis include the age of the individual (each period is a year) starting at 20.

   (c) (10 points) Now do the same simulation for 10,000 individuals with the same starting condition $a = 0$ and "blue-collar" worker income. Note that the path for income should be different every time. As a consequence, the paths for assets and consumption will be different too. Plot in the y axis the mean of consumption, income, and assets for each age and age in the x axis. Do the same plot for the variance of log consumption and log income.

   (d) (15 points) Again with your favorite method, solve for $a'(a, y)$ but setting $\underline{a} = -50$. Do the same simulation you did in (c) and plot in the y axis the mean of consumption, income, and assets for each age and age in the x axis. Do the same plot for

the variance of log consumption and log income. If you plot together the cases of $\underline{a} = 0$ $\underline{a} = -50$ your figure up to age 60 should look like (not very close, but with some similar features) this one in Kaplan and Violante (American Economic Journal: Macroeconomics, 2010).

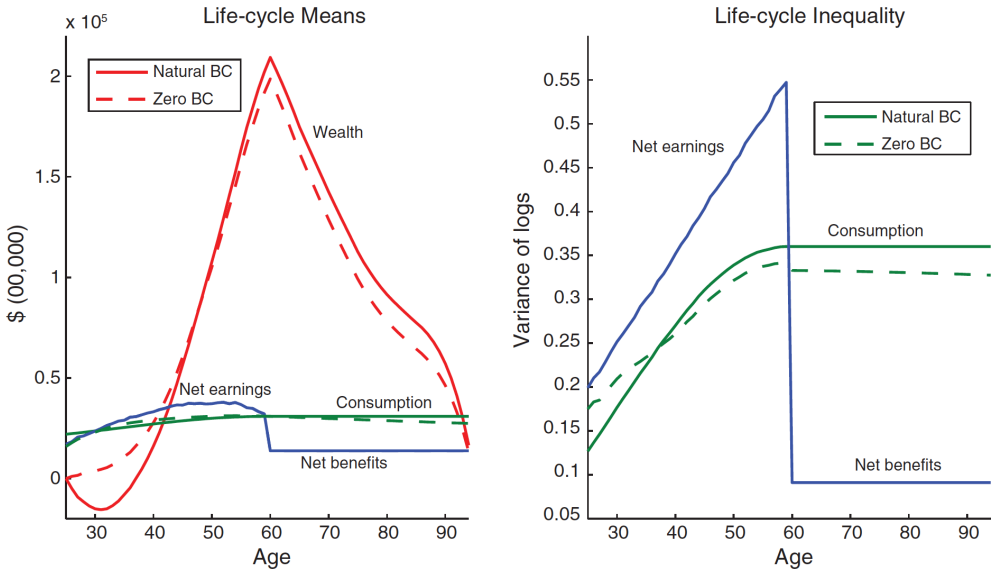


FIGURE 1. LIFE-CYCLE PROFILES FOR MEANS AND VARIANCES IN THE NBC AND ZBC ECONOMIES

(e) (15 points) Kapan and Violante also compute the "insurance coefficients" implied by the model. The insurance coefficient for income shocks in our framework are

$$\phi^{\Delta y} = 1 - \frac{cov(\Delta y_{i,t}, \Delta c_{i,t})}{var(\Delta y_{i,t})} \tag{1}$$

If consumption and income changes co-variate perfectly, there is no insurance. Read their paper to learn more about how you should compute the insurance coefficient. You can compute this covariance and variance with your simulated data and obtain $\phi^{\Delta y}$. What value do you obtain for the cases of $\underline{a} = 0$ and $\underline{a} = -50$? How does it compare with Kaplan and Violante (2010)'s finding (Model TRUE is the one you are computing).

TABLE 1—RESULTS FROM THE BENCHMARK MODELS WITH NBC AND ZBC

	Permanent shock			Transitory shock		
	Data BPP	Model BPP	Model TRUE	Data BPP	Model BPP	Model TRUE
Natural BC	0.36 (0.09)	0.22	0.23	0.95 (0.04)	0.94	0.94
Zero BC	0.36 (0.09)	0.07	0.23	0.95 (0.04)	0.82	0.82

(f) (20 points) For the case of $\underline{a} = -50$, keeping $R = 1.02$, and using the method of discretization and interpolation of the density function f discussed in class. Plot an histogram of age, z, ε, and assets. What is the aggregate stock of assets in the economy?

Note that among other parameters, this stock of assets depends on the interest rate R, so we will write $A(R)$. In this case what you found is $A(1.02)$.

(g) (20 points) Solve the policy function $a'(a, y)$ and the density function f for $R = 1.04$. Compute the aggregate stocks of assets for this new level R, $A(1.04)$. Do the same for $R = 0.98$, $R = 1.0$, and $R = 1.07$. Note that now you have found $A(0.98), A(1.0), A(1.02), A(1.04), A(1.07)$. Plot this with R in the x axis and $A(R)$ in the y axis. What would you guess is the value of R such that $A(R) = 0$? Why?